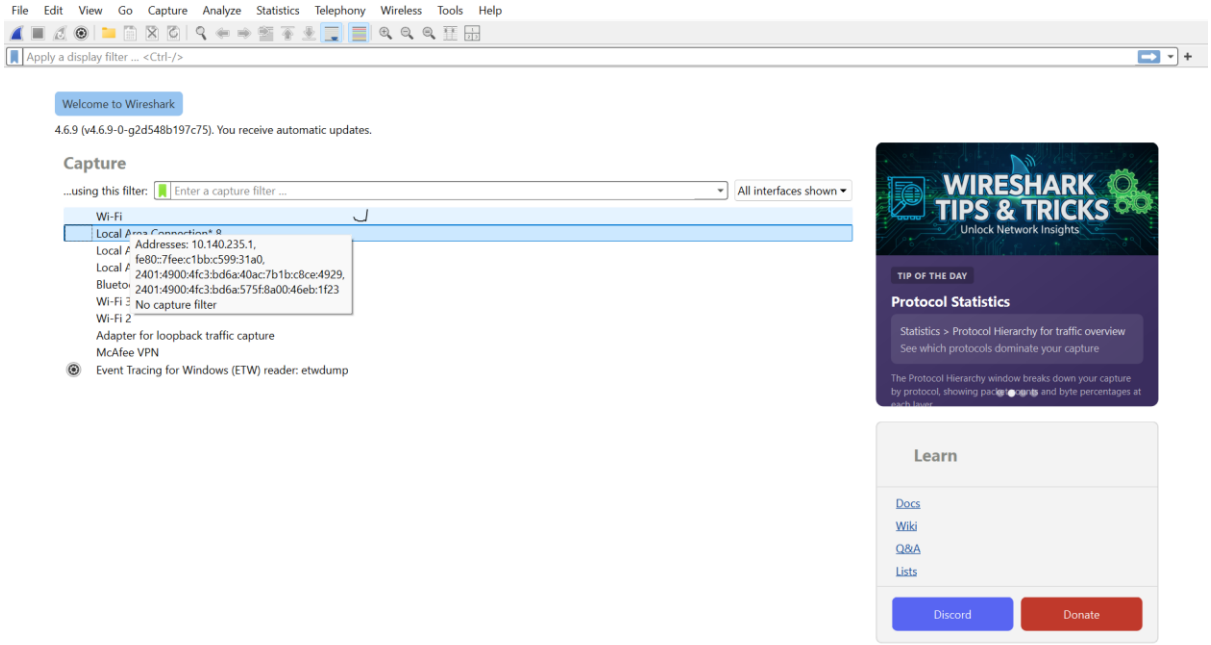
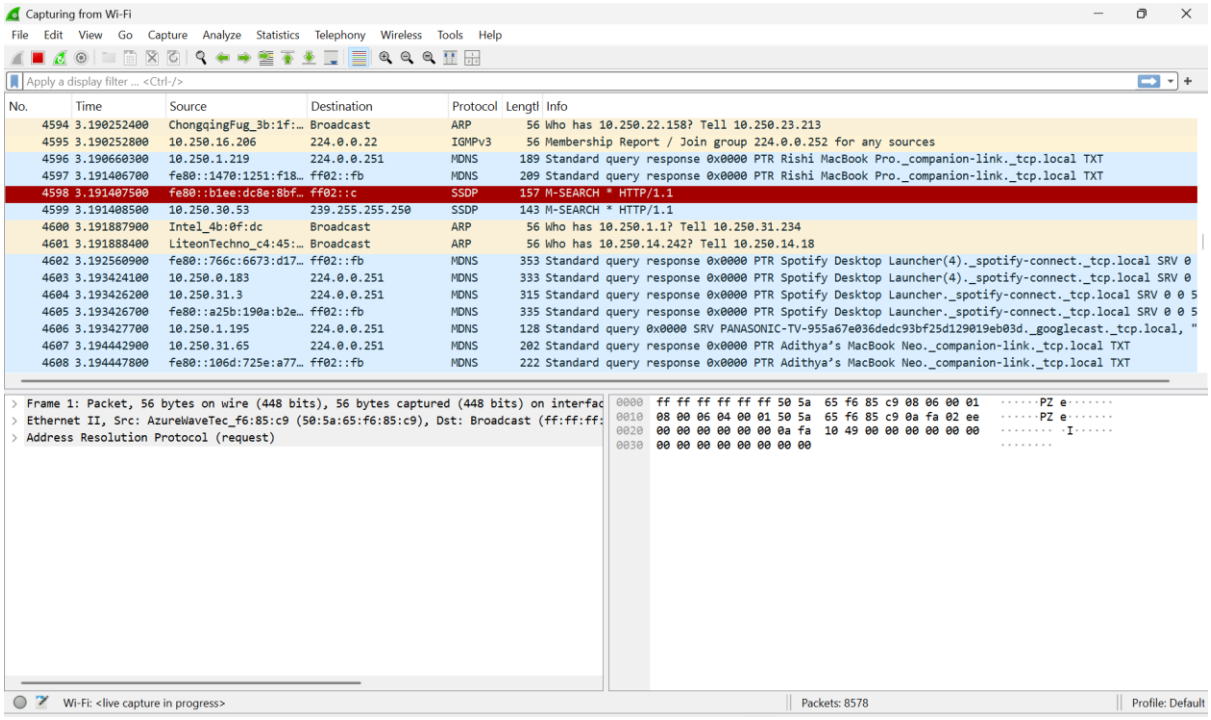
WEEK -1

STEP-1: Installisation of wireshark

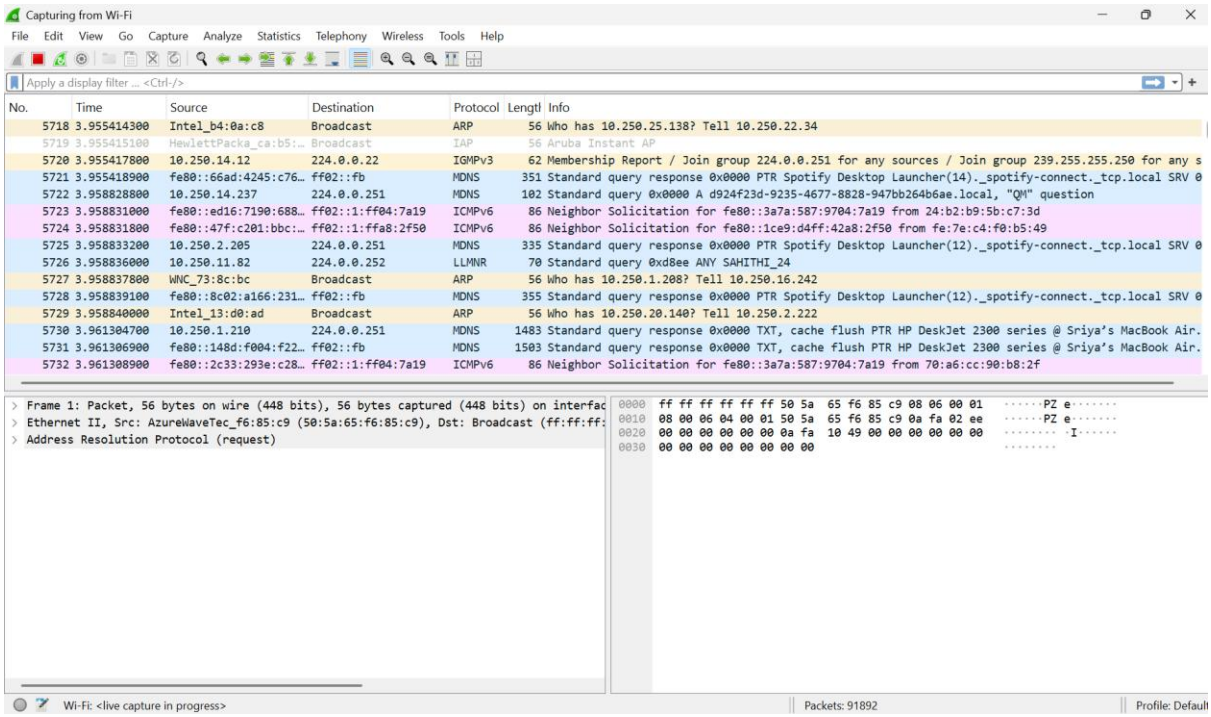
STEP-2:



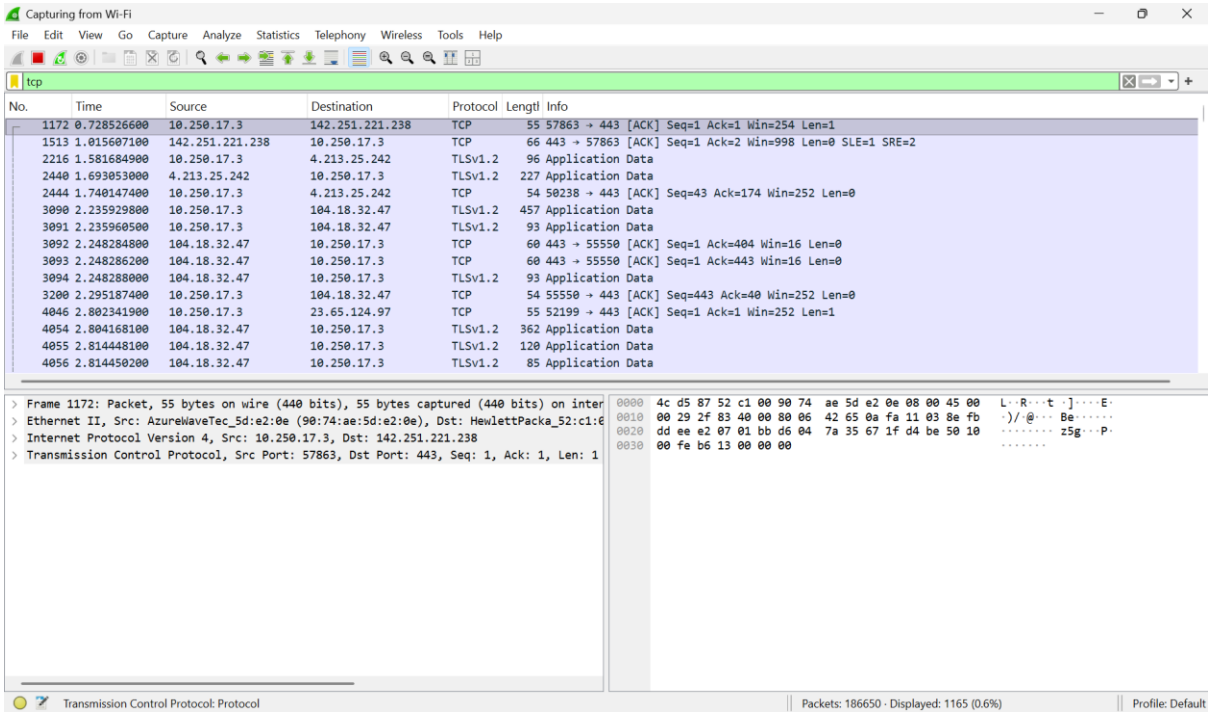
STEP-3:

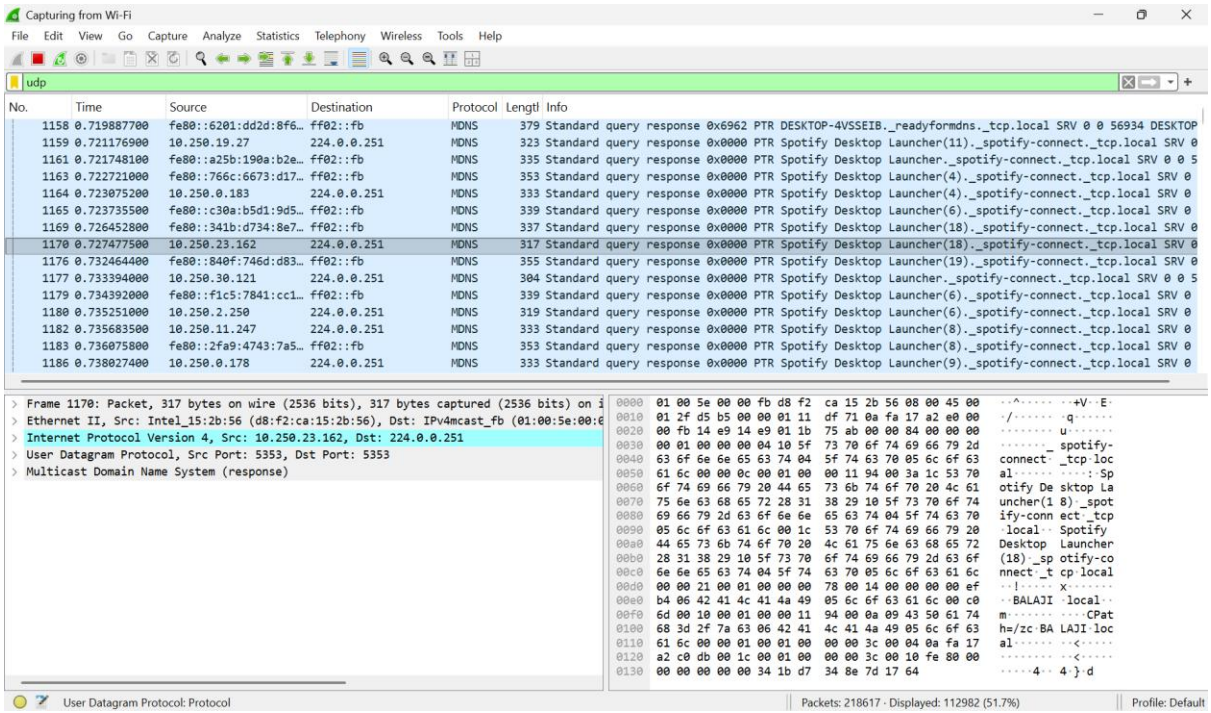


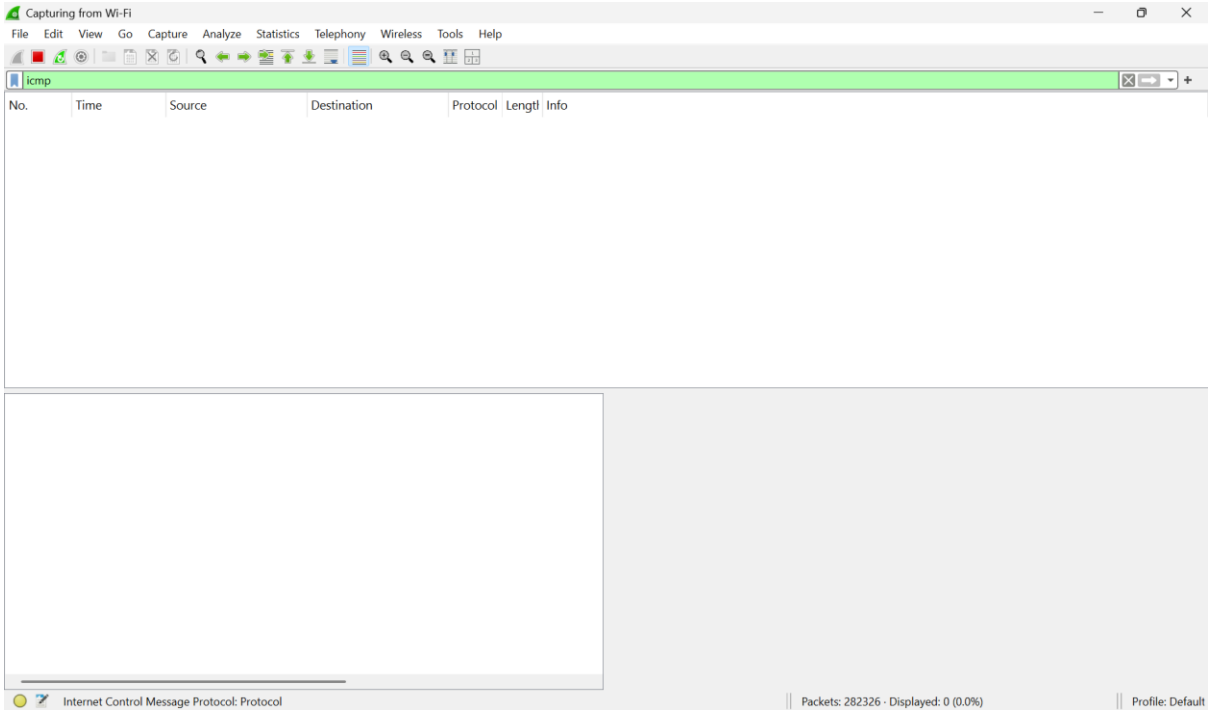
STEP-4:

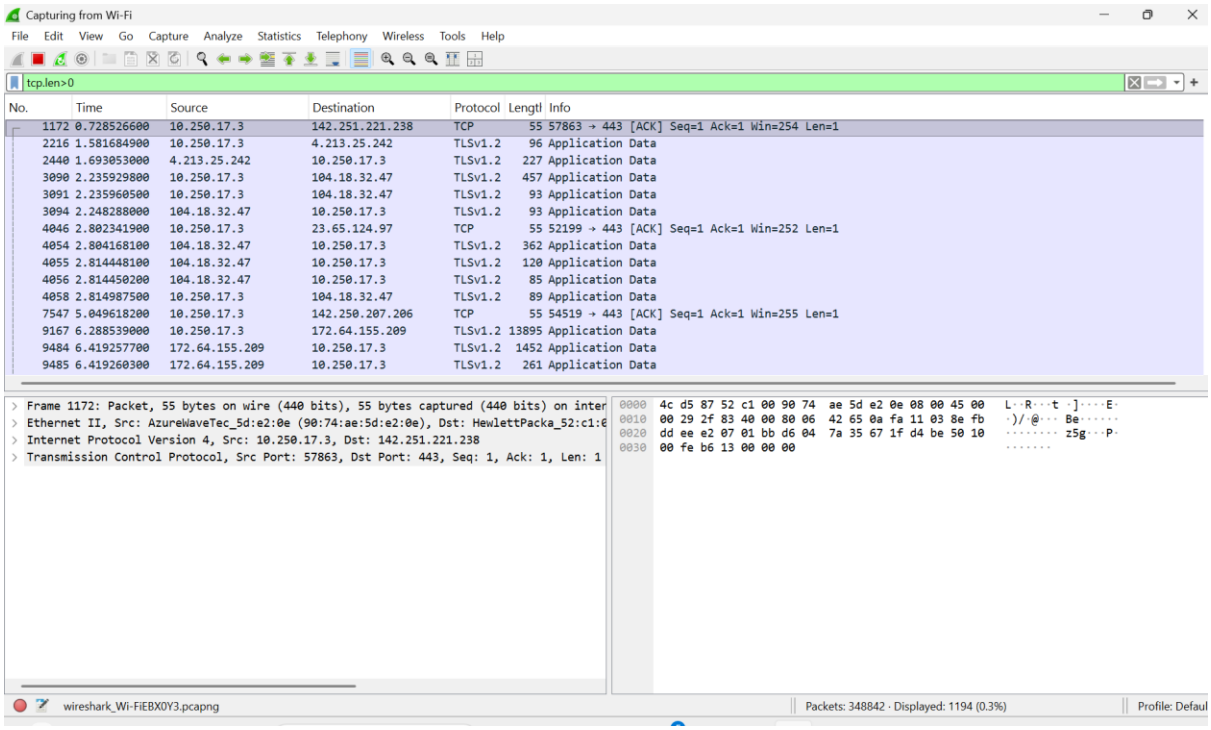

Capturing from Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
No. Time Source Destination Protocol Lengtl Info
5718 3.955414300 Intel_b4:0a:c8 Broadcast ARP 56 Who has 10.250.25.138? Tell 10.250.22.34
5719 3.955415100 HewlettPacka_ca:b5:... Broadcast IAP 56 Aruba Instant AP
5720 3.955417800 10.250.14.12 224.0.0.22 IGMPv3 62 Membership Report / Join group 224.0.0.251 for any sources / Join group 239.255.255.250 for any s
5721 3.955418900 fe80::66ad:4245:c76... ff02::fb MDNS 351 Standard query response 0x0000 PTR Spotify Desktop Launcher(14)._spotify-connect._tcp.local SRV 0
5722 3.958828800 10.250.14.237 224.0.0.251 MDNS 102 Standard query 0x0000 A d924f23d-9235-4677-8828-947bb264b6ae.local, "QM" question
5723 3.958831000 fe80::ed16:7190:688... ff02::1:ff04:7a19 ICMPv6 86 Neighbor Solicitation for fe80::3a7a:587:9704:7a19 from 24:b2:b9:5b:c7:3d
5724 3.958831800 fe80::47f:c201:bbc:... ff02::1:ffa8:2f50 ICMPv6 86 Neighbor Solicitation for fe80::1ce9:d4ff:42a8:2f50 from fe:7e:c4:f0:b5:49
5725 3.958833200 10.250.2.205 224.0.0.251 MDNS 335 Standard query response 0x0000 PTR Spotify Desktop Launcher(12)._spotify-connect._tcp.local SRV 0
5726 3.958836000 10.250.11.82 224.0.0.252 LLMNR 70 Standard query 0xd8ee ANY SAHITHI_24
5727 3.958837800 WNC_73:8c:bc Broadcast ARP 56 Who has 10.250.1.208? Tell 10.250.16.242
5728 3.958839100 fe80::8c02:a166:231... ff02::fb MDNS 355 Standard query response 0x0000 PTR Spotify Desktop Launcher(12)._spotify-connect._tcp.local SRV 0
5729 3.958840000 Intel_13:d0:ad Broadcast ARP 56 Who has 10.250.20.140? Tell 10.250.2.222
5730 3.961304700 10.250.1.210 224.0.0.251 MDNS 1483 Standard query response 0x0000 TXT, cache flush PTR HP DeskJet 2300 series @ Sriya's MacBook Air.
5731 3.961306900 fe80::148d:f004:f22... ff02::fb MDNS 1503 Standard query response 0x0000 TXT, cache flush PTR HP DeskJet 2300 series @ Sriya's MacBook Air.
5732 3.961308900 fe80::2c33:293e:c28... ff02::1:ff04:7a19 ICMPv6 86 Neighbor Solicitation for fe80::3a7a:587:9704:7a19 from 70:a6:cc:90:b8:2f
Frame 1: Packet, 56 bytes on wire (448 bits), 56 bytes captured (448 bits) on interfac
Ethernet II, Src: AzureWaveTec_f6:85:c9 (50:5a:65:f6:85:c9), Dst: Broadcast (ff:ff:ff:
Address Resolution Protocol (request)
0000 ff ff ff ff ff ff 50 5a 65 f6 85 c9 08 06 00 01 ·····PZ e·····
0010 08 00 06 04 00 01 50 5a 65 f6 85 c9 0a fa 02 ee ·····PZ e·····
0020 00 00 00 00 00 00 0a fa 10 49 00 00 00 00 00 00 ·····I·····
0030 00 00 00 00 00 00 00 00 ····
Wi-Fi: <live capture in progress>
Packets: 91892
Profile: Default

STEP-5:


Capturing from Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp
No. Time Source Destination Protocol Lengtl Info
1172 0.728526600 10.250.17.3 142.251.221.238 TCP 55 57863 → 443 [ACK] Seq=1 Ack=1 Win=254 Len=1
1513 1.015607100 142.251.221.238 10.250.17.3 TCP 66 443 → 57863 [ACK] Seq=1 Ack=2 Win=998 Len=0 SLE=1 SRE=2
2216 1.581684900 10.250.17.3 4.213.25.242 TLSv1.2 96 Application Data
2440 1.693053000 4.213.25.242 10.250.17.3 TLSv1.2 227 Application Data
2444 1.740147400 10.250.17.3 4.213.25.242 TCP 54 50238 → 443 [ACK] Seq=43 Ack=174 Win=252 Len=0
3090 2.235929800 10.250.17.3 104.18.32.47 TLSv1.2 457 Application Data
3091 2.235960500 10.250.17.3 104.18.32.47 TLSv1.2 93 Application Data
3092 2.248284800 104.18.32.47 10.250.17.3 TCP 60 443 → 55550 [ACK] Seq=1 Ack=404 Win=16 Len=0
3093 2.248286200 104.18.32.47 10.250.17.3 TCP 60 443 → 55550 [ACK] Seq=1 Ack=443 Win=16 Len=0
3094 2.248288000 104.18.32.47 10.250.17.3 TLSv1.2 93 Application Data
3200 2.295187400 10.250.17.3 104.18.32.47 TCP 54 55550 → 443 [ACK] Seq=443 Ack=40 Win=252 Len=0
4046 2.802341900 10.250.17.3 23.65.124.97 TCP 55 52199 → 443 [ACK] Seq=1 Ack=1 Win=252 Len=1
4054 2.804168100 104.18.32.47 10.250.17.3 TLSv1.2 362 Application Data
4055 2.814448100 104.18.32.47 10.250.17.3 TLSv1.2 120 Application Data
4056 2.814450200 104.18.32.47 10.250.17.3 TLSv1.2 85 Application Data
Frame 1172: Packet, 55 bytes on wire (440 bits), 55 bytes captured (440 bits) on inter
Ethernet II, Src: AzureWaveTec_5d:e2:0e (90:74:ae:5d:e2:0e), Dst: HewlettPacka_52:c1:6
Internet Protocol Version 4, Src: 10.250.17.3, Dst: 142.251.221.238
Transmission Control Protocol, Src Port: 57863, Dst Port: 443, Seq: 1, Ack: 1, Len: 1
0000 4c d5 87 52 c1 00 90 74 ae 5d e2 0e 08 00 45 00 L··R···t ·]····E·
0010 00 29 2f 83 40 00 80 06 42 65 0a fa 11 03 8e fb ·)/·@··· Be·····
0020 dd ee e2 07 01 bb d6 04 7a 35 67 1f d4 be 50 10 ····· z5g···P·
0030 00 fe b6 13 00 00 00 ····
Transmission Control Protocol: Protocol
Packets: 186650 · Displayed: 1165 (0.6%)
Profile: Default

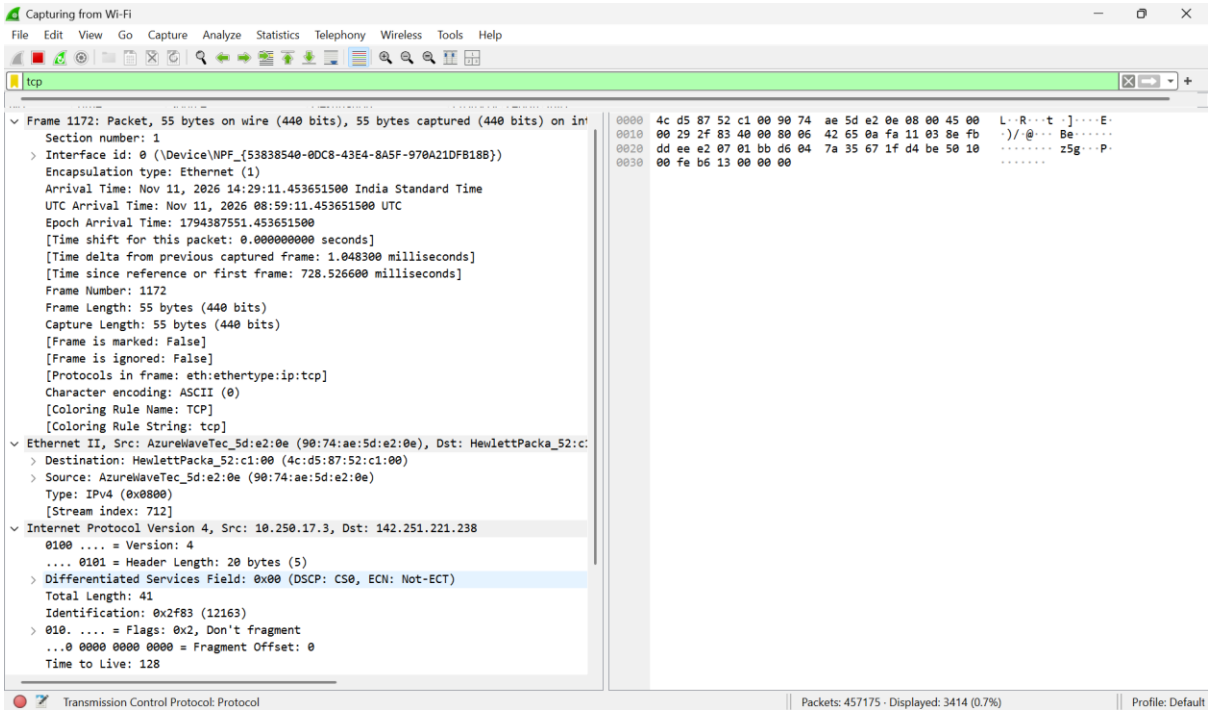

Capturing from Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
udp
No. Time Source Destination Protocol Lengtl Info
1158 0.719887700 fe80::6201:dd2d:8f6… ff02::fb MDNS 379 Standard query response 0x6962 PTR DESKTOP-4VSSEIB._readyformdns._tcp.local SRV 0 0 56934 DESKTOP
1159 0.721176900 10.250.19.27 224.0.0.251 MDNS 323 Standard query response 0x0000 PTR Spotify Desktop Launcher(11)._spotify-connect._tcp.local SRV 0
1161 0.721748100 fe80::a25b:190a:b2e… ff02::fb MDNS 335 Standard query response 0x0000 PTR Spotify Desktop Launcher._spotify-connect._tcp.local SRV 0 0 5
1163 0.722721000 fe80::766c:6673:d17… ff02::fb MDNS 353 Standard query response 0x0000 PTR Spotify Desktop Launcher(4)._spotify-connect._tcp.local SRV 0
1164 0.723075200 10.250.0.183 224.0.0.251 MDNS 333 Standard query response 0x0000 PTR Spotify Desktop Launcher(4)._spotify-connect._tcp.local SRV 0
1165 0.723735500 fe80::c30a:b5d1:9d5… ff02::fb MDNS 339 Standard query response 0x0000 PTR Spotify Desktop Launcher(6)._spotify-connect._tcp.local SRV 0
1169 0.726452800 fe80::341b:d734:8e7… ff02::fb MDNS 337 Standard query response 0x0000 PTR Spotify Desktop Launcher(18)._spotify-connect._tcp.local SRV 0
1170 0.727477500 10.250.23.162 224.0.0.251 MDNS 317 Standard query response 0x0000 PTR Spotify Desktop Launcher(18)._spotify-connect._tcp.local SRV 0
1176 0.732464400 fe80::840f:746d:d83… ff02::fb MDNS 355 Standard query response 0x0000 PTR Spotify Desktop Launcher(19)._spotify-connect._tcp.local SRV 0
1177 0.733394000 10.250.30.121 224.0.0.251 MDNS 304 Standard query response 0x0000 PTR Spotify Desktop Launcher._spotify-connect._tcp.local SRV 0 0 5
1179 0.734392000 fe80::f1c5:7841:cc1… ff02::fb MDNS 339 Standard query response 0x0000 PTR Spotify Desktop Launcher(6)._spotify-connect._tcp.local SRV 0
1180 0.735251000 10.250.2.250 224.0.0.251 MDNS 319 Standard query response 0x0000 PTR Spotify Desktop Launcher(6)._spotify-connect._tcp.local SRV 0
1182 0.735683500 10.250.11.247 224.0.0.251 MDNS 333 Standard query response 0x0000 PTR Spotify Desktop Launcher(8)._spotify-connect._tcp.local SRV 0
1183 0.736075800 fe80::2fa9:4743:7a5… ff02::fb MDNS 353 Standard query response 0x0000 PTR Spotify Desktop Launcher(8)._spotify-connect._tcp.local SRV 0
1186 0.738027400 10.250.0.178 224.0.0.251 MDNS 333 Standard query response 0x0000 PTR Spotify Desktop Launcher(9)._spotify-connect._tcp.local SRV 0
Frame 1170: Packet, 317 bytes on wire (2536 bits), 317 bytes captured (2536 bits) on i
Ethernet II, Src: Intel_15:2b:56 (d8:f2:ca:15:2b:56), Dst: IPv4mcast_fb (01:00:5e:00:0
Internet Protocol Version 4, Src: 10.250.23.162, Dst: 224.0.0.251
User Datagram Protocol, Src Port: 5353, Dst Port: 5353
Multicast Domain Name System (response)
0000 01 00 5e 00 00 fb d8 f2 ca 15 2b 56 08 00 45 00
0010 01 2f d5 b5 00 00 01 11 df 71 0a fa 17 a2 e0 00
0020 00 fb 14 e9 14 e9 01 1b 75 ab 00 00 84 00 00 00
0030 00 01 00 00 00 04 10 5f 73 70 6f 74 69 66 79 2d
0040 63 6f 6e 6e 65 63 74 04 5f 74 63 70 05 6c 6f 63
0050 61 6c 00 00 0c 00 01 00 00 11 94 00 3a 1c 53 70
0060 6f 74 69 66 79 20 44 65 73 6b 74 6f 70 20 4c 61
0070 75 6e 63 68 65 72 28 31 38 29 10 5f 73 70 6f 74
0080 69 66 79 2d 63 6f 6e 6e 65 63 74 04 5f 74 63 70
0090 05 6c 6f 63 61 6c 00 1c 53 70 6f 74 69 66 79 20
00a0 44 65 73 6b 74 6f 70 20 4c 61 75 6e 63 68 65 72
00b0 28 31 38 29 10 5f 73 70 6f 74 69 66 79 2d 63 6f
00c0 6e 6e 65 63 74 04 5f 74 63 70 05 6c 6f 63 61 6c
00d0 00 00 21 00 01 00 00 00 78 00 14 00 00 00 00 ef
00e0 b4 06 42 41 4c 41 4a 49 05 6c 6f 63 61 6c 00 c0
00f0 6d 00 10 00 01 00 00 11 94 00 0a 09 43 50 61 74
0100 68 3d 2f 7a 63 06 42 41 4c 41 4a 49 05 6c 6f 63
0110 61 6c 00 00 01 00 01 00 00 00 3c 00 04 0a fa 17
0120 a2 c0 db 00 1c 00 01 00 00 00 3c 00 10 fe 80 00
0130 00 00 00 00 00 34 1b d7 34 8e 7d 17 64
User Datagram Protocol: Protocol
Packets: 218617 · Displayed: 112982 (51.7%)
Profile: Default

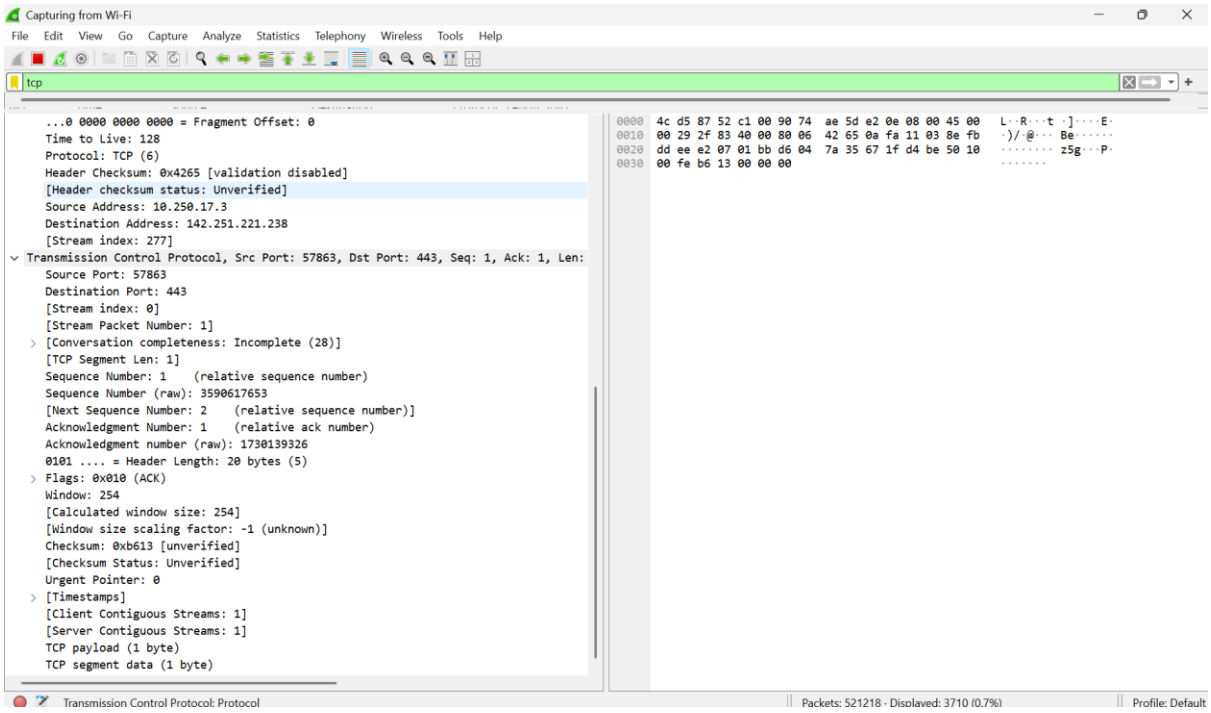

Capturing from Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
icmp
No. Time Source Destination Protocol Lengtl Info
Internet Control Message Protocol: Protocol
Packets: 282326 · Displayed: 0 (0.0%)
Profile: Default


Capturing from Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp.len>0
No. Time Source Destination Protocol Lengtl Info
1172 0.728526600 10.250.17.3 142.251.221.238 TCP 55 57863 → 443 [ACK] Seq=1 Ack=1 Win=254 Len=1
2216 1.581684900 10.250.17.3 4.213.25.242 TLSv1.2 96 Application Data
2440 1.693053000 4.213.25.242 10.250.17.3 TLSv1.2 227 Application Data
3090 2.235929800 10.250.17.3 104.18.32.47 TLSv1.2 457 Application Data
3091 2.235960500 10.250.17.3 104.18.32.47 TLSv1.2 93 Application Data
3094 2.248288000 104.18.32.47 10.250.17.3 TLSv1.2 93 Application Data
4046 2.802341900 10.250.17.3 23.65.124.97 TCP 55 52199 → 443 [ACK] Seq=1 Ack=1 Win=252 Len=1
4054 2.804168100 104.18.32.47 10.250.17.3 TLSv1.2 362 Application Data
4055 2.814448100 104.18.32.47 10.250.17.3 TLSv1.2 120 Application Data
4056 2.814450200 104.18.32.47 10.250.17.3 TLSv1.2 85 Application Data
4058 2.814987500 10.250.17.3 104.18.32.47 TLSv1.2 89 Application Data
7547 5.049618200 10.250.17.3 142.250.207.206 TCP 55 54519 → 443 [ACK] Seq=1 Ack=1 Win=255 Len=1
9167 6.288539000 10.250.17.3 172.64.155.209 TLSv1.2 13895 Application Data
9484 6.419257700 172.64.155.209 10.250.17.3 TLSv1.2 1452 Application Data
9485 6.419260300 172.64.155.209 10.250.17.3 TLSv1.2 261 Application Data
Frame 1172: Packet, 55 bytes on wire (440 bits), 55 bytes captured (440 bits) on inter
Ethernet II, Src: AzureWaveTec_5d:e2:0e (90:74:ae:5d:e2:0e), Dst: HewlettPacka_52:c1:0
Internet Protocol Version 4, Src: 10.250.17.3, Dst: 142.251.221.238
Transmission Control Protocol, Src Port: 57863, Dst Port: 443, Seq: 1, Ack: 1, Len: 1
0000 4c d5 87 52 c1 00 90 74 ae 5d e2 0e 08 00 45 00
0010 00 29 2f 83 40 00 80 06 42 65 0a fa 11 03 8e fb
0020 dd ee e2 07 01 bb d6 04 7a 35 67 1f d4 be 50 10
0030 00 fe b6 13 00 00 00
wireshark_Wi-FiEBX0Y3.pcapng
Packets: 348842 · Displayed: 1194 (0.3%)
Profile: Defaul

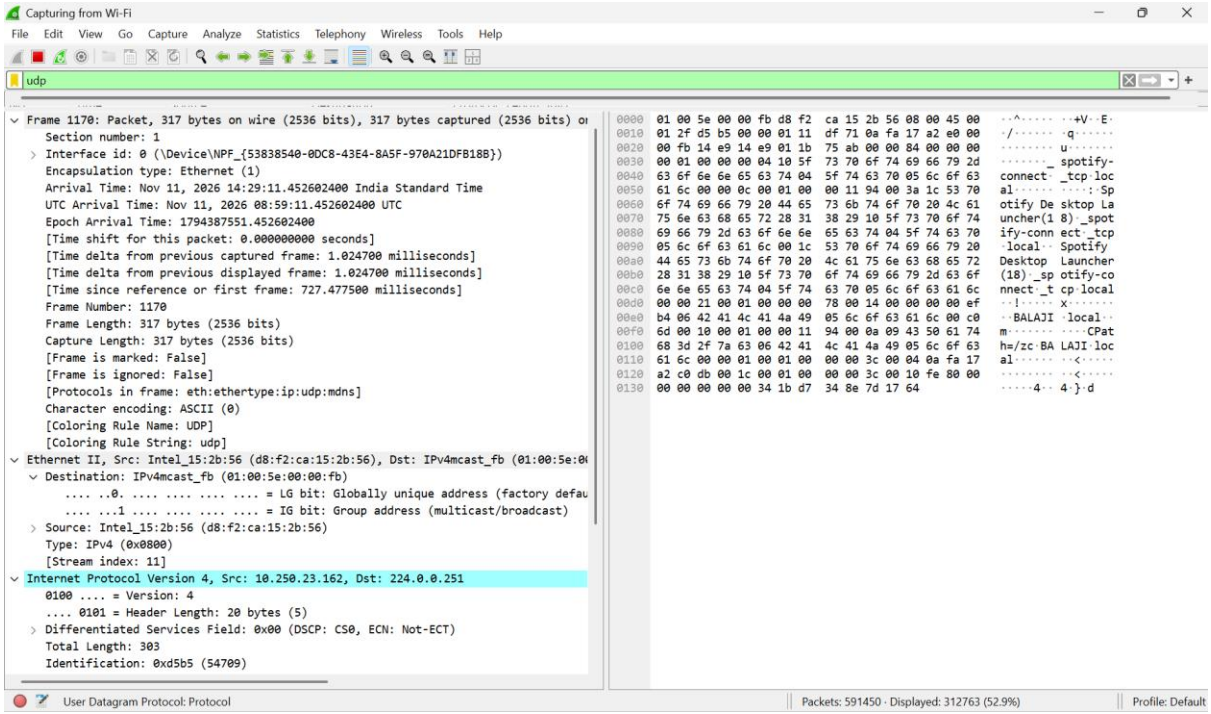
STEP-6:

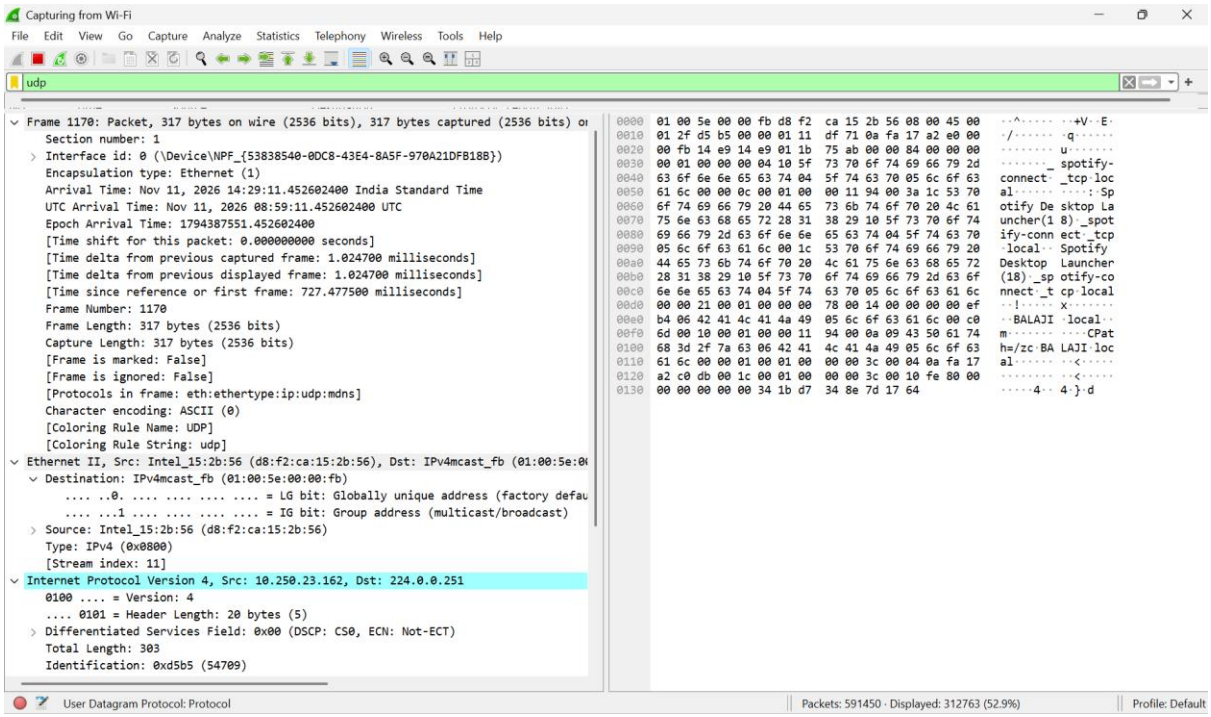
TCP


Capturing from Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp
Frame 1172: Packet, 55 bytes on wire (440 bits), 55 bytes captured (440 bits) on int
Section number: 1
Interface id: 0 (\Device\NPF_{53838540-0DC8-43E4-8A5F-970A21DFB18B})
Encapsulation type: Ethernet (1)
Arrival Time: Nov 11, 2026 14:29:11.453651500 India Standard Time
UTC Arrival Time: Nov 11, 2026 08:59:11.453651500 UTC
Epoch Arrival Time: 1794387551.453651500
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 1.048300 milliseconds]
[Time since reference or first frame: 728.526600 milliseconds]
Frame Number: 1172
Frame Length: 55 bytes (440 bits)
Capture Length: 55 bytes (440 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ip:tcp]
Character encoding: ASCII (0)
[Coloring Rule Name: TCP]
[Coloring Rule String: tcp]
Ethernet II, Src: AzureWaveTec_5d:e2:0e (90:74:ae:5d:e2:0e), Dst: HewlettPacka_52:c1
Destination: HewlettPacka_52:c1:00 (4c:d5:87:52:c1:00)
Source: AzureWaveTec_5d:e2:0e (90:74:ae:5d:e2:0e)
Type: IPv4 (0x0800)
[Stream index: 712]
Internet Protocol Version 4, Src: 10.250.17.3, Dst: 142.251.221.238
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 41
Identification: 0x2f83 (12163)
010. .... = Flags: 0x2, Don't fragment
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 128
0000 4c d5 87 52 c1 00 90 74 ae 5d e2 0e 08 00 45 00
0010 00 29 2f 83 40 00 80 06 42 65 0a fa 11 03 8e fb
0020 dd ee e2 07 01 bb d6 04 7a 35 67 1f d4 be 50 10
0030 00 fe b6 13 00 00 00
Transmission Control Protocol: Protocol
Packets: 457175 · Displayed: 3414 (0.7%)
Profile: Default

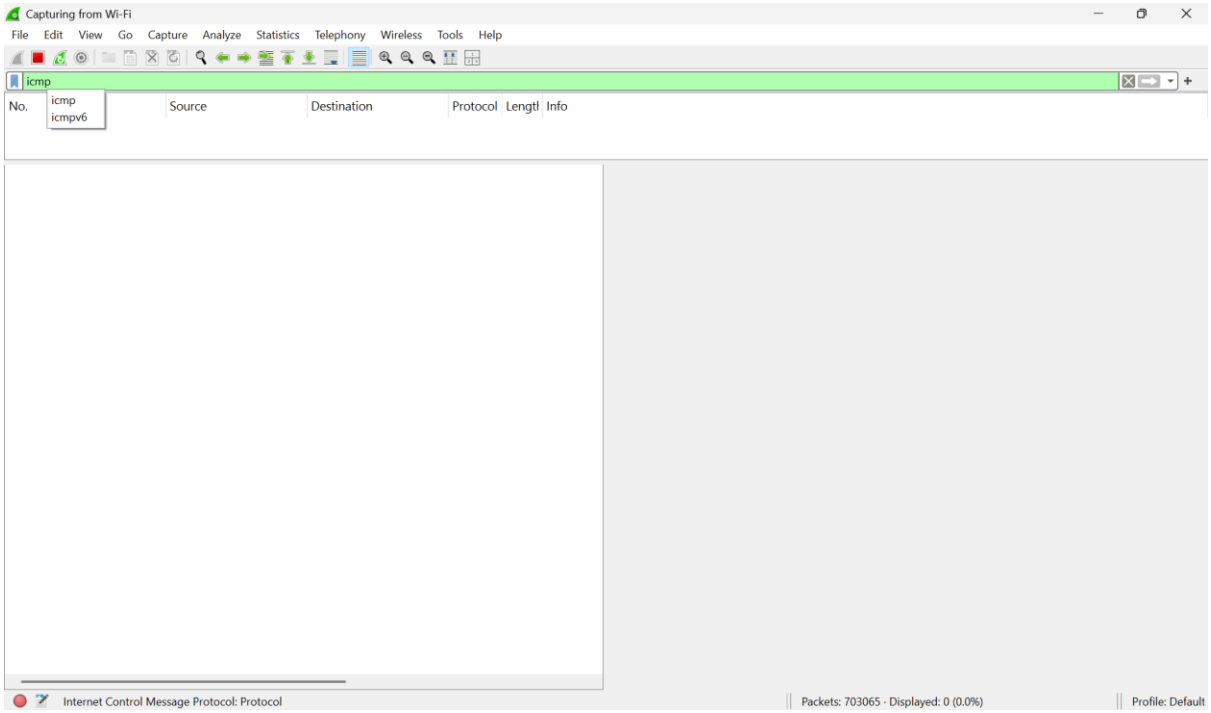

Capturing from Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 128
Protocol: TCP (6)
Header Checksum: 0x4265 [validation disabled]
[Header checksum status: Unverified]
Source Address: 10.250.17.3
Destination Address: 142.251.221.238
[Stream index: 277]
Transmission Control Protocol, Src Port: 57863, Dst Port: 443, Seq: 1, Ack: 1, Len:
Source Port: 57863
Destination Port: 443
[Stream index: 0]
[Stream Packet Number: 1]
[Conversation completeness: Incomplete (28)]
[TCP Segment Len: 1]
Sequence Number: 1 (relative sequence number)
Sequence Number (raw): 3590617653
[Next Sequence Number: 2 (relative sequence number)]
Acknowledgment Number: 1 (relative ack number)
Acknowledgment number (raw): 1730139326
0101 .... = Header Length: 20 bytes (5)
Flags: 0x010 (ACK)
Window: 254
[Calculated window size: 254]
[Window size scaling factor: -1 (unknown)]
Checksum: 0xb613 [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
[Timestamps]
[Client Contiguous Streams: 1]
[Server Contiguous Streams: 1]
TCP payload (1 byte)
TCP segment data (1 byte)
0000 4c d5 87 52 c1 00 90 74 ae 5d e2 0e 08 00 45 00
0010 00 29 2f 83 40 00 80 06 42 65 0a fa 11 03 8e fb
0020 dd ee e2 07 01 bb d6 04 7a 35 67 1f d4 be 50 10
0030 00 fe b6 13 00 00 00
Transmission Control Protocol: Protocol
Packets: 521218 · Displayed: 3710 (0.7%)
Profile: Default

UDP

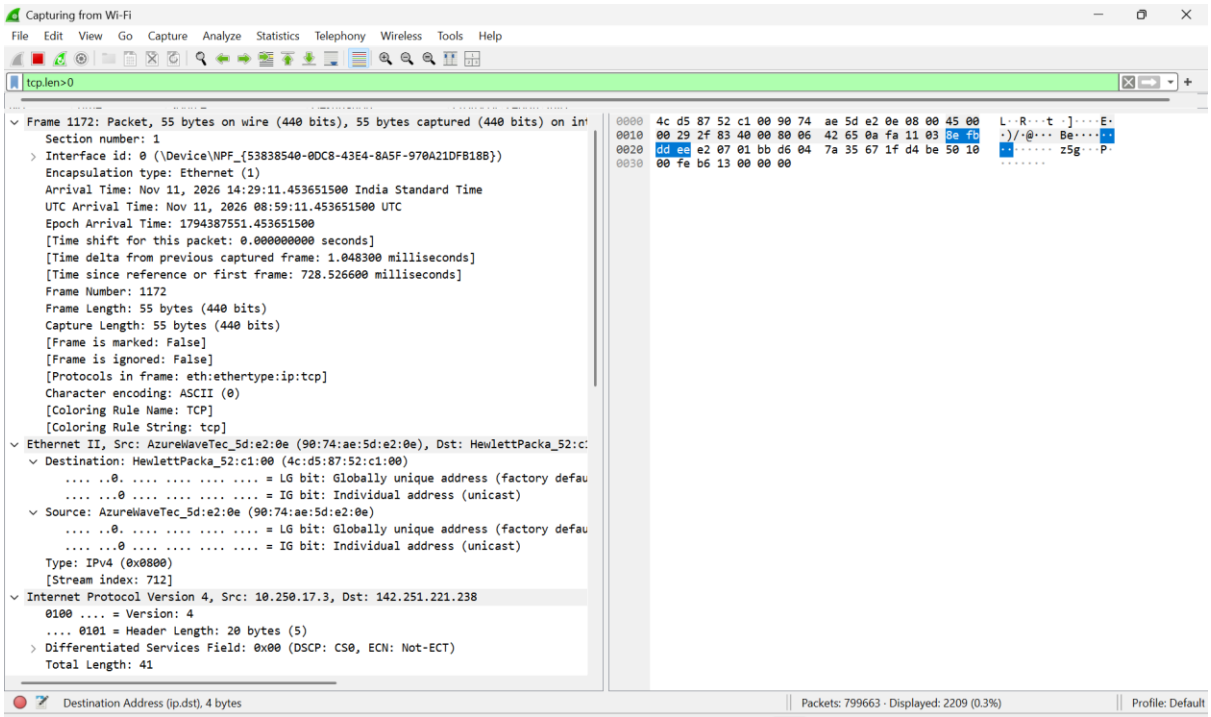

Capturing from Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
udp
Frame 1170: Packet, 317 bytes on wire (2536 bits), 317 bytes captured (2536 bits) or
Section number: 1
Interface id: 0 (\Device\NPF_{53838540-0DC8-43E4-8A5F-970A21DFB18B})
Encapsulation type: Ethernet (1)
Arrival Time: Nov 11, 2026 14:29:11.452602400 India Standard Time
UTC Arrival Time: Nov 11, 2026 08:59:11.452602400 UTC
Epoch Arrival Time: 1794387551.452602400
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 1.024700 milliseconds]
[Time delta from previous displayed frame: 1.024700 milliseconds]
[Time since reference or first frame: 727.477500 milliseconds]
Frame Number: 1170
Frame Length: 317 bytes (2536 bits)
Capture Length: 317 bytes (2536 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ip:udp:mdns]
Character encoding: ASCII (0)
[Coloring Rule Name: UDP]
[Coloring Rule String: udp]
Ethernet II, Src: Intel_15:2b:56 (d8:f2:ca:15:2b:56), Dst: IPv4mcast_fb (01:00:5e:0
Destination: IPv4mcast_fb (01:00:5e:00:00:fb)
.... ..0. .... = LG bit: Globally unique address (factory defau
....1 .... = IG bit: Group address (multicast/broadcast)
Source: Intel_15:2b:56 (d8:f2:ca:15:2b:56)
Type: IPv4 (0x0800)
[Stream index: 11]
Internet Protocol Version 4, Src: 10.250.23.162, Dst: 224.0.0.251
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 303
Identification: 0xd5b5 (54709)
0000 01 00 5e 00 00 fb d8 f2 ca 15 2b 56 08 00 45 00
0010 01 2f d5 b5 00 00 01 11 df 71 0a fa 17 a2 e0 00
0020 00 fb 14 e9 14 e9 01 1b 75 ab 00 00 84 00 00 00
0030 00 01 00 00 00 04 10 5f 73 70 6f 74 69 66 79 2d
0040 63 6f 6e 6e 65 63 74 04 5f 74 63 70 05 6c 6f 63
0050 61 6c 00 00 0c 00 01 00 00 11 94 00 3a 1c 53 70
0060 6f 74 69 66 79 20 44 65 73 6b 74 6f 70 20 4c 61
0070 75 6e 63 68 65 72 28 31 38 29 10 5f 73 70 6f 74
0080 69 66 79 2d 63 6f 6e 6e 65 63 74 04 5f 74 63 70
0090 05 6c 6f 63 61 6c 00 1c 53 70 6f 74 69 66 79 20
00a0 44 65 73 6b 74 6f 70 20 4c 61 75 6e 63 68 65 72
00b0 28 31 38 29 10 5f 73 70 6f 74 69 66 79 2d 63 6f
00c0 6e 6e 65 63 74 04 5f 74 63 70 05 6c 6f 63 61 6c
00d0 00 00 21 00 01 00 00 00 78 00 14 00 00 00 00 ef
00e0 b4 06 42 41 4c 41 4a 49 05 6c 6f 63 61 6c 00 c0
00f0 6d 00 10 00 01 00 00 11 94 00 0a 09 43 50 61 74
0100 68 3d 2f 7a 63 06 42 41 4c 41 4a 49 05 6c 6f 63
0110 61 6c 00 00 01 00 01 00 00 00 3c 00 04 0a fa 17
0120 a2 c0 db 00 1c 00 01 00 00 00 3c 00 10 fe 80 00
0130 00 00 00 00 00 34 1b d7 34 8e 7d 17 64
User Datagram Protocol: Protocol
Packets: 591450 · Displayed: 312763 (52.9%)
Profile: Default

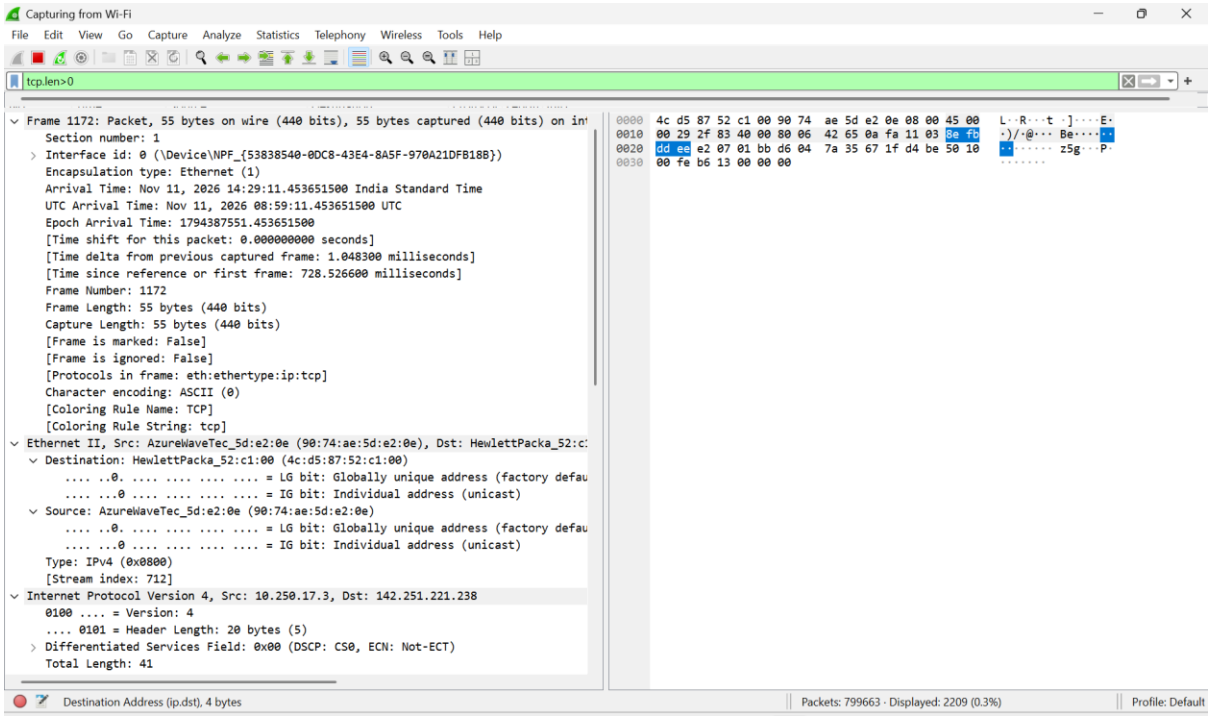


ICMP



TCP.LEN>0


Capturing from Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp.len>0
Frame 1172: Packet, 55 bytes on wire (440 bits), 55 bytes captured (440 bits) on int
Section number: 1
Interface id: 0 (\Device\NPF_{53838540-0DC8-43E4-8A5F-970A21DFB18B})
Encapsulation type: Ethernet (1)
Arrival Time: Nov 11, 2026 14:29:11.453651500 India Standard Time
UTC Arrival Time: Nov 11, 2026 08:59:11.453651500 UTC
Epoch Arrival Time: 1794387551.453651500
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 1.048300 milliseconds]
[Time since reference or first frame: 728.526600 milliseconds]
Frame Number: 1172
Frame Length: 55 bytes (440 bits)
Capture Length: 55 bytes (440 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ip:tcp]
Character encoding: ASCII (0)
[Coloring Rule Name: TCP]
[Coloring Rule String: tcp]
Ethernet II, Src: AzureWaveTec_5d:e2:0e (90:74:ae:5d:e2:0e), Dst: HewlettPacka_52:c1
Destination: HewlettPacka_52:c1:00 (4c:d5:87:52:c1:00)
.... ..0. .... = LG bit: Globally unique address (factory defau
....0 .... = IG bit: Individual address (unicast)
Source: AzureWaveTec_5d:e2:0e (90:74:ae:5d:e2:0e)
.... ..0. .... = LG bit: Globally unique address (factory defau
....0 .... = IG bit: Individual address (unicast)
Type: IPv4 (0x0800)
[Stream index: 712]
Internet Protocol Version 4, Src: 10.250.17.3, Dst: 142.251.221.238
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 41
0000 4c d5 87 52 c1 00 90 74 ae 5d e2 0e 08 00 45 00
0010 00 29 2f 83 40 00 80 06 42 65 0a fa 11 03 8e fb
0020 dd ee e2 07 01 bb d6 04 7a 35 67 1f d4 be 50 10
0030 00 fe b6 13 00 00 00
Destination Address (ip.dst), 4 bytes
Packets: 799663 · Displayed: 2209 (0.3%)
Profile: Default


Capturing from Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp.len>0
Frame 1172: Packet, 55 bytes on wire (440 bits), 55 bytes captured (440 bits) on int
Section number: 1
Interface id: 0 (\Device\NPF_{53838540-0DC8-43E4-8A5F-970A21DFB18B})
Encapsulation type: Ethernet (1)
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[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 1.048300 milliseconds]
[Time since reference or first frame: 728.526600 milliseconds]
Frame Number: 1172
Frame Length: 55 bytes (440 bits)
Capture Length: 55 bytes (440 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ip:tcp]
Character encoding: ASCII (0)
[Coloring Rule Name: TCP]
[Coloring Rule String: tcp]
Ethernet II, Src: AzureWaveTec_5d:e2:0e (90:74:ae:5d:e2:0e), Dst: HewlettPacka_52:c1
Destination: HewlettPacka_52:c1:00 (4c:d5:87:52:c1:00)
.... ..0. .... = LG bit: Globally unique address (factory defau
....0 .... = IG bit: Individual address (unicast)
Source: AzureWaveTec_5d:e2:0e (90:74:ae:5d:e2:0e)
.... ..0. .... = LG bit: Globally unique address (factory defau
....0 .... = IG bit: Individual address (unicast)
Type: IPv4 (0x0800)
[Stream index: 712]
Internet Protocol Version 4, Src: 10.250.17.3, Dst: 142.251.221.238
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 41
0000 4c d5 87 52 c1 00 90 74 ae 5d e2 0e 08 00 45 00
0010 00 29 2f 83 40 00 80 06 42 65 0a fa 11 03 8e fb
0020 dd ee e2 07 01 bb d6 04 7a 35 67 1f d4 be 50 10
0030 00 fe b6 13 00 00 00
Destination Address (ip.dst), 4 bytes
Packets: 799663 · Displayed: 2209 (0.3%)
Profile: Default


[Checksum Status: Unverified]
Urgent Pointer: 0
[Timestamps]
[Client Contiguous Streams: 1]
[Server Contiguous Streams: 1]
TCP payload (1 byte)
TCP segment data (1 byte)
Destination Address (ip.dst), 4 bytes
Packets: 848510 · Displayed: 2336 (0.3%)
Profile: Default

STEP-7:

1. TCP Packet

From my Wireshark capture:

Parameter	**Value**
Frame Length	55 bytes
Ethernet Header	14 bytes
IPv4 Header	20 bytes
TCP Header	20 bytes
Application Data	1 byte
FCS	Not included

Total Size:

$$14 + 20 + 20 + 1 = 55 \text{ bytes}$$

Total Header Overhead:

$$14 + 20 + 20 = 54 \text{ bytes}$$

Payload Efficiency:

$$\frac{1}{55} \times 100 = 1.82\%$$

2. UDP Packet

From my Wireshark capture:

Parameter	**Value**
Frame Length	317 bytes
Ethernet Header	14 bytes
IPv4 Header	20 bytes
UDP Header	8 bytes
Application Data	275 bytes

FCS	Not included

Total Size:

$$14 + 20 + 8 + 275 = 317 \text{ bytes}$$

Total Header Overhead:

$$14 + 20 + 8 = 42 \text{ bytes}$$

Payload Efficiency:

$$\frac{275}{317} \times 100 = 86.75\%$$

STEP-8:

Based on your Wireshark captured values:

Parameter	**TCP Packet**	**UDP Packet**
Frame Length	55 bytes	317 bytes
Ethernet Header Length	14 bytes	14 bytes
IP Header Length	20 bytes(IPv4)	20 bytes (IPv4)
Transport Header Length	20 bytes(TCP)	8 bytes (UDP)
IP Total Length	41 bytes	303 bytes
Payload Length	1 byte	275 bytes